

# Understanding Mathematical Concepts via Go Language

Region

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# 고 언어로 수학적 개념 이해하기

영역

강효상 (DGIST)

**Region**

# Set and Region

```
type Set[T Element] interface {  
    Contains(T) bool  
}  
  
type Region interface {  
    Set[Point]  
    Corner() Vector  
    Lengths() []Real  
}
```

# Implementation of Region - Interval

```
type Interval [2]float64

func (i Interval) Contains(p gocalc.Point) bool {
    if p.Len() != 1 {
        panic("Interval.Contains: P dim mismatch")
    }
    x := p.Map(0).ToFloat()
    return i[0] <= x && x <= i[1]
}

func (i Interval) Corner() gocalc.Vector {
    return SimpleVector{i[0]}
}

func (i Interval) Lengths() []gocalc.Real {
    return []gocalc.Real{SimpleReal(i[1] - i[0])}
}
```

# Implementation of Region - Rectangular Region

```
type Rect []Interval

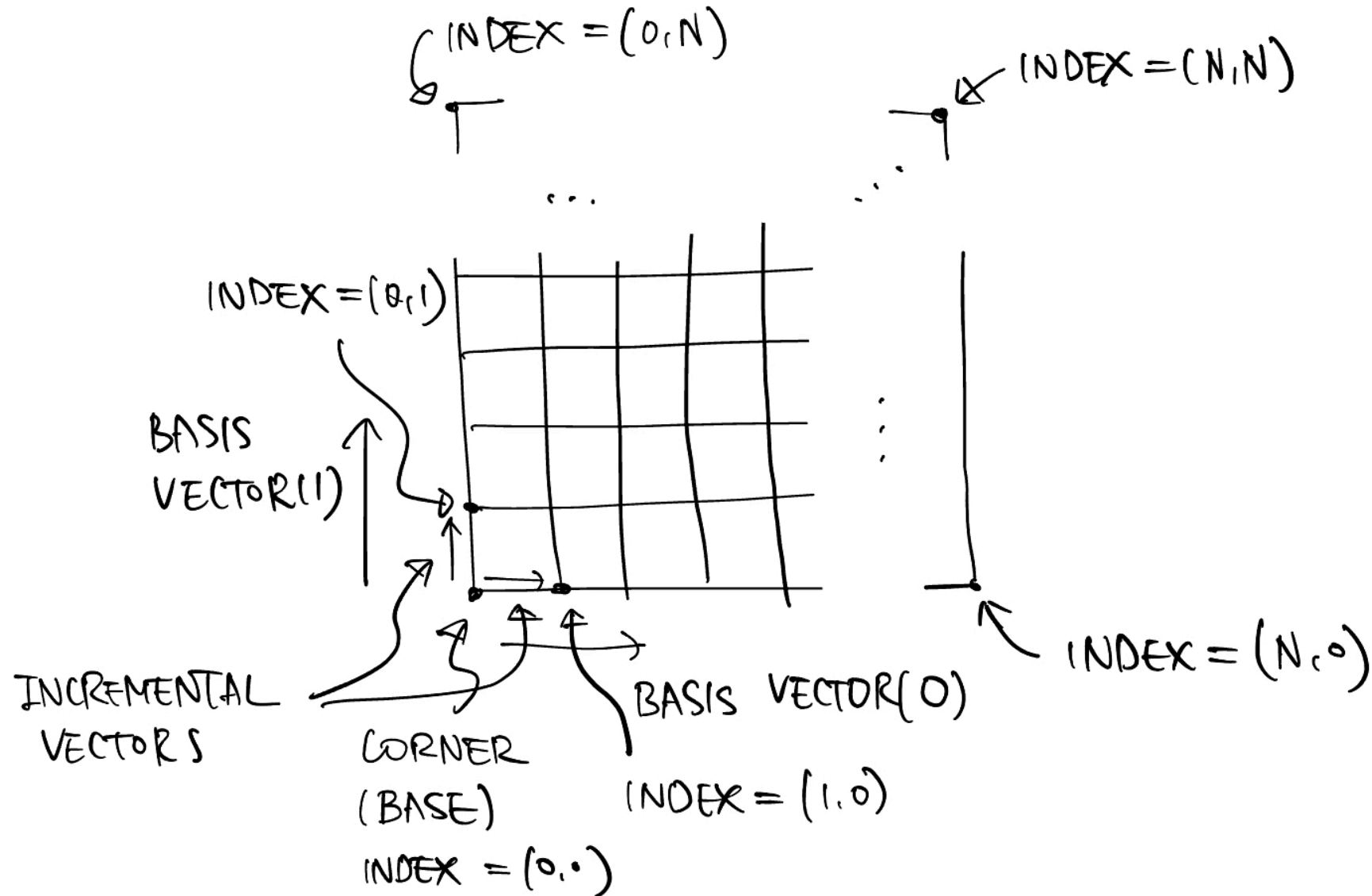
var _ gocalc.Region = Rect{}

func (r Rect) Contains(p gocalc.Point) bool {
    for i := 0; i < len(r); i++ {
        x := p.Map(i).ToFloat()
        if r[i][0] > x || x > r[i][1] {
            return false
        }
    }
    return true
}

func (r Rect) Corner() gocalc.Vector {
    v := make(SimpleVector, len(r))
    for i := 0; i < len(r); i++ {
        v[i] = r[i][0]
    }
    return v
}

func (r Rect) Lengths() []gocalc.Real {
    lengths := make([]gocalc.Real, len(r))
    for i := 0; i < len(r); i++ {
        lengths[i] = SimpleReal(r[i][1] - r[i][0])
    }
    return lengths
}
```

# Subdivision of Region



```

func Mesh(reg Region, nsub int) []Vector {
    base := reg.Corner()
    basis := base.Basis()
    lengths := reg.Lengths()
    if len(lengths) != len(basis) {
        panic("Mesh: lengths and basis have different dimensions")
    }
    var bdry []Vector
    for i, b := range basis {
        bb := b.Scale(lengths[i].Mul(Rational(base.Map(0), [2]int{1, nsub})).(Real))
        bdry = append(bdry, bb)
    }
    var mesh []Vector
    for v := range SubInc(bdry, nsub) {
        mesh = append(mesh, base.Add(v).(Vector))
    }
    return mesh
}

```



```

func SubInc(vds []Vector, n int) <-chan Vector {
    if len(vds) < 1 || n < 1 {
        panic("SubInc: Wrong number(s).")
    }
    ch := make(chan Vector)
    go func() {
        defer close(ch)
        p := vds[0].Zero().(Vector)
        for i := 0; i <= n; i++ {
            if len(vds) == 1 {
                ch <- p
            } else {
                for v := range SubInc(vds[1:], n) {
                    ch <- p.Add(v).(Vector)
                }
            }
            p = p.Add(vds[0]).(Vector)
        }
    }()
    return ch
}

```

**This video was produced with the support of CTL in DGIST.**

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